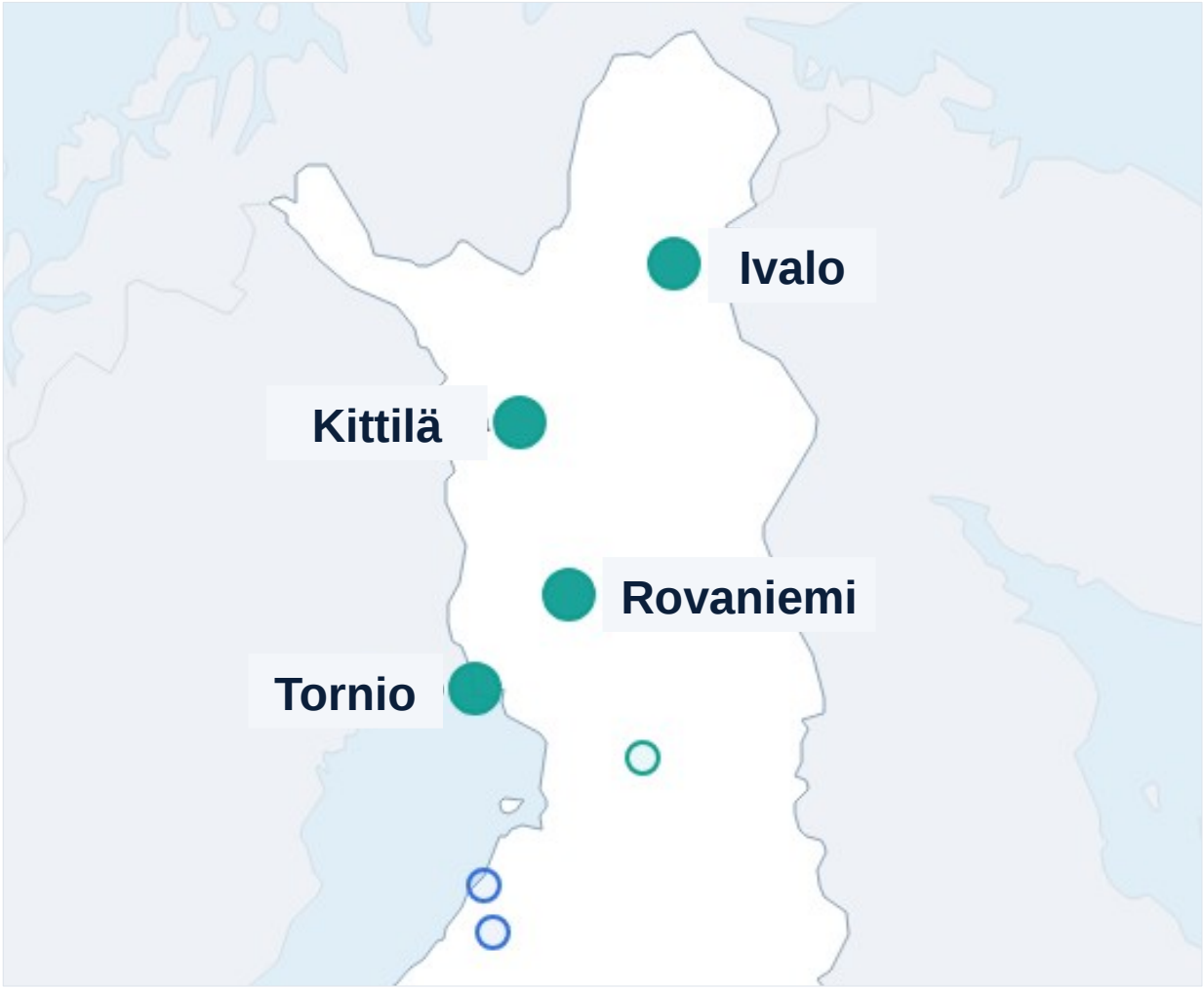
In May 2005, four Lapland rivers hit flood levels together.

Lapland's snow had warned us two months earlier.

Flow over 1-in-10 level · snow top 15 % · in-sample



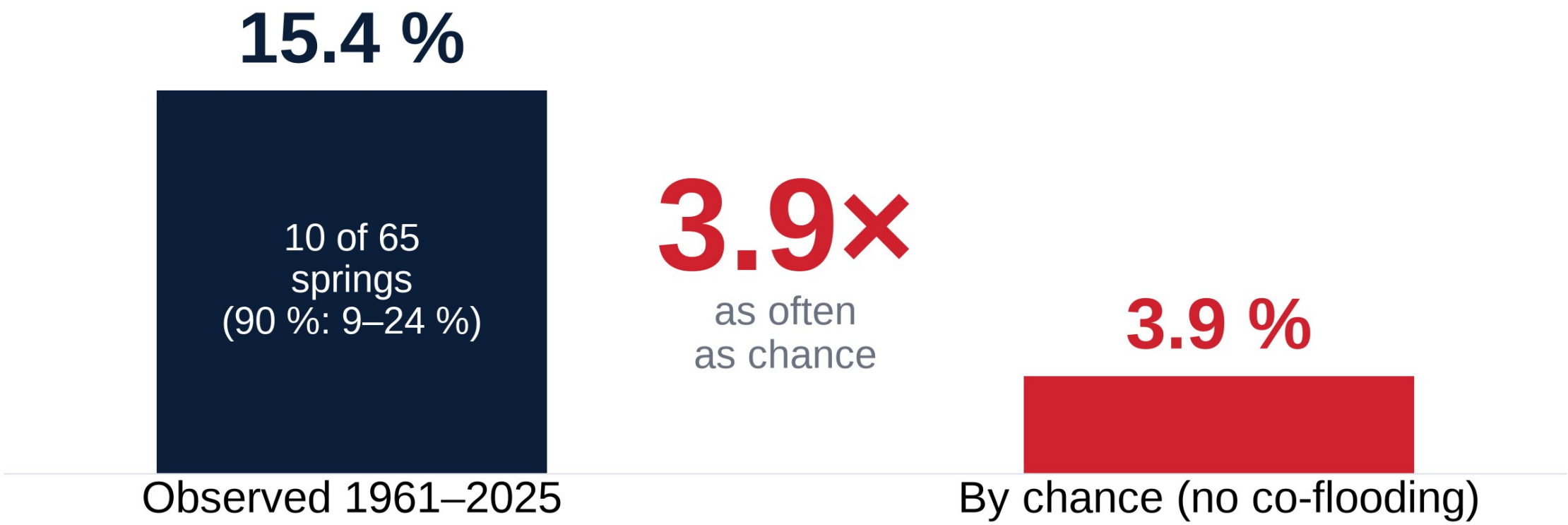
Tulva app, spring 2005: filled = over its 1-in-10 flood level

Source: SYKE discharge 1961–2025 and 1 March snow (SWE)

THE GAP

Finland's rivers flood together, not one at a time

Springs when ≥4 of 14 river areas pass their 1-in-10 flood level



Cover is bought in autumn. The snow signal comes in March.

Source: SYKE 1961–2025 (+ SMHI Torne); chance = rivers flood independently, same gauge-years as observed

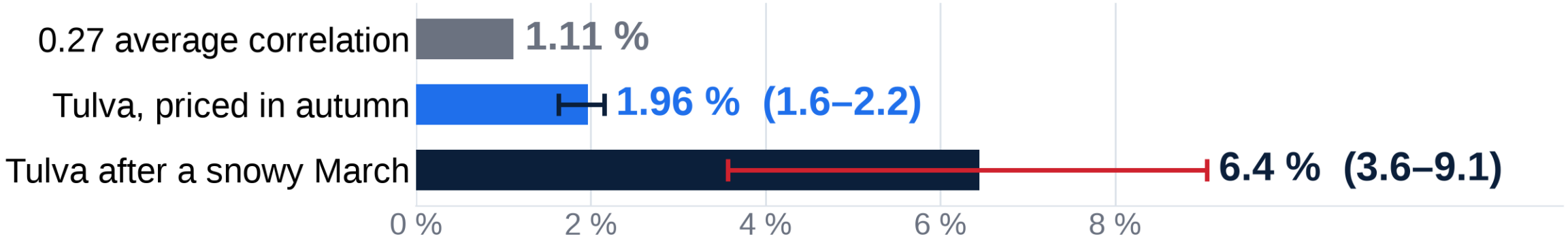
THE PRODUCT

Tulva Index: sold in autumn, updated in March

Pays the full limit if ≥2 of 5 northern rivers pass their 1-in-50 level

Expected loss ≈1.8× a model with the 2027 matrix's average correlation (0.27)

EXPECTED LOSS, % OF THE LIMIT PER YEAR (90 % INTERVALS)



After a snowy March, expected loss triples vs autumn (1.96 % → 6.4 %; 90 %: 2.1–4.4×)
Priced in autumn. In March the insurer acts on it: top-up cover, claims staff in the regions at risk.

Sources: SYKE 1961–2025, Tulva model, year-bootstrap intervals; 0.27 = mean of the 19-zone matrix, Del. Reg. (EU) 2026/269

Tulva

Joint risk | Snow outlook | Index cover | How it works | Under the hood | Year replay

1 · JOINT RISK

Finland's spring floods don't arrive one at a time.

The 14 river flood-risk areas fall into three clusters that share a snowpack. Lines join areas whose spring peaks move together (Spearman ρ ≥ 0.6, 1961–2025).

2.8×

per river pair: as often as chance, two rivers of one cluster both over 1-in-10 in the same spring (pairs) EXACT

10 of 65

per spring: springs with ≥4 of 14 areas over their 1-in-10 level (3.9× chance) EXACT

- North · 5 areas · 2.9× joint floods Pudasjärvi, Ivalo, Kittilä, Rovaniemi, Tornio
- Ostrobothnia · 6 areas · 3.2× joint floods Lapväärtti, Ilmajoki–Seinäjoki, Laihia, Ylistaro–Koivulahti, Alavieska–Ylivieska, Lower Pyhäjoki
- Satakunta–SE · 3 areas · 1.3× joint floods Pori, Huittinen, Lower Kymijoki



Our app: lines join rivers that flood together

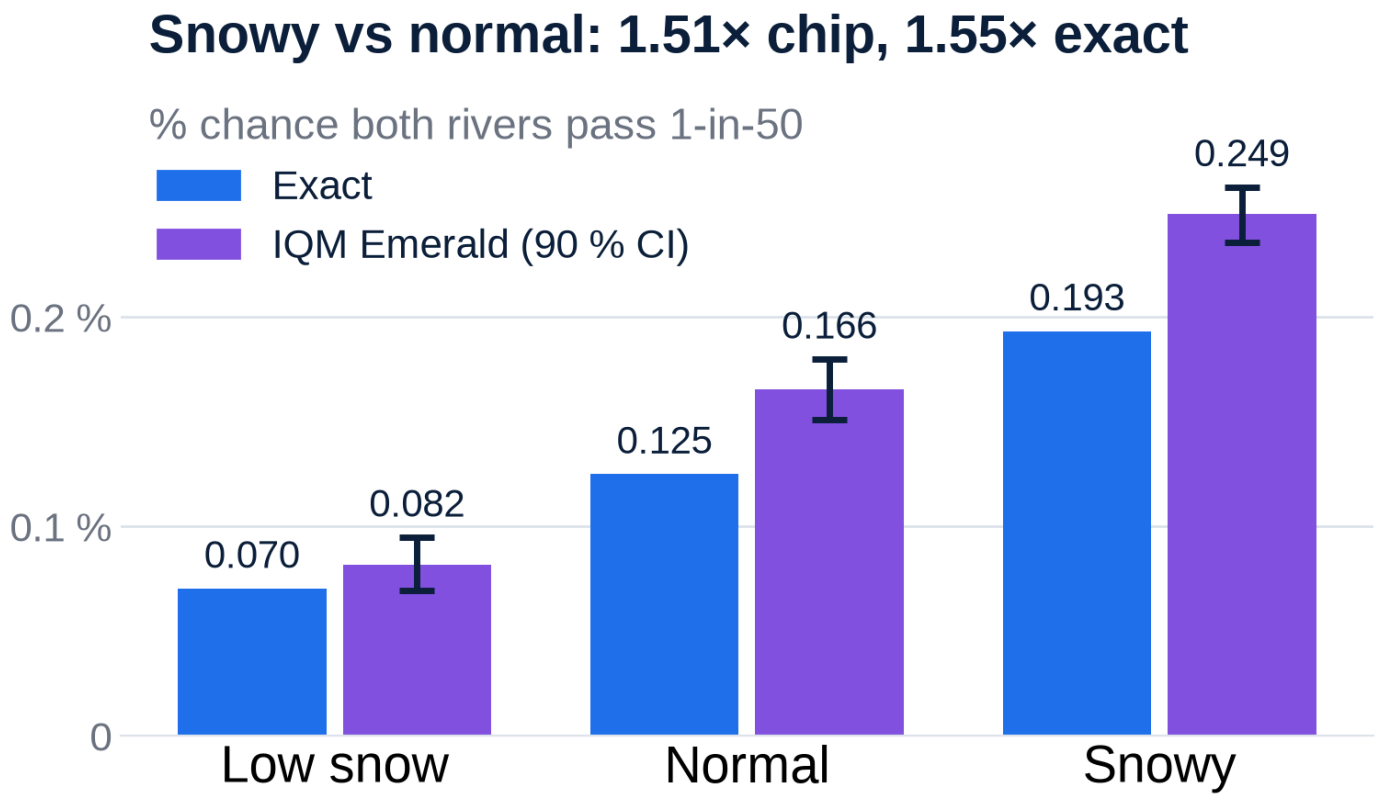
Our app: joint-flood map → March snow → quantum chip → 2005 replay

QUANTUM, IN PRACTICE

It runs on four real quantum computers today

Snowy vs low-snow March: 3.05× on chip, 2.75× exact

Two-river North-cluster pair, IQM Emerald



CALIBRATED SAMPLING + SHALLOW AMPLITUDE ESTIMATION

Two-river core

IBM Marrakesh, IQM Garnet and Emerald, VTT Q50: 3 qubits

Simplified 14-river model

Two-level version, 18 qubits, sampling only: IQM Emerald and VTT Q50

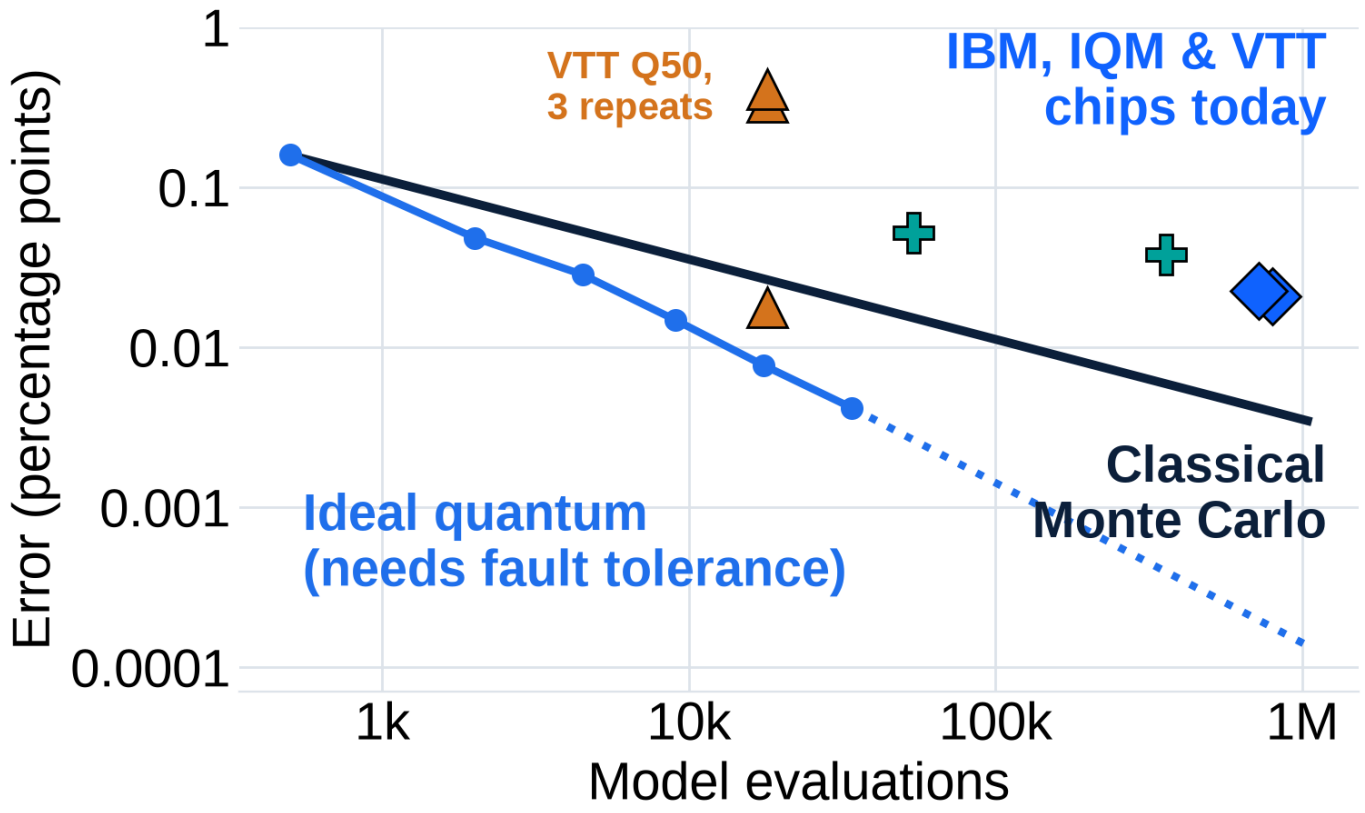
VTT Q50, Finland's own

3 repeats: 0.11–0.53 % vs 0.125 % exact. Calibration not yet stable

Intervals, amplification, raw readings and job IDs: appendix

QUANTUM, HONESTLY

Classical Monte Carlo still wins today



Two-river North-cluster pair, both over 1-in-50: error vs effort

Method ready, hardware not yet

Today: rare floods drown in noise.

Prices are computed classically; quantum is the R&D track.

14-river chip runs and the noise control: appendix

BUSINESS

Who pays whom, and why now

TULVA

Index + pricing licence

€25k pilot, then ~€75k a year per insurer

← licence

BUYER

Finnish home insurers

OP, If, LähiTapiola, …

→ premium

RISK CARRIER

A reinsurer

≈€0.5–0.7M a year per €10M layer

YEAR 3

3–6 insurers × €75k ≈ €0.2–0.45M/yr

Upside: Nordic snowmelt (assumption)

Why now From 2027 the regulator's standard formula fixes flood correlations that ignore the snow; insurers with their own approved (internal) model, like If, can use a snow-aware joint-flood number.

Basis risk 1-in-10 back-test: paid in loss springs 1966, 2005, 2018; missed Ostrobothnia 2012–13, 2024 and Lapland 2020; since 2005 also paid in 3 North springs with no recorded loss (2015, 2023, 2024).

Ask One insurer's flood claims for 8 weeks → basis-risk map + firm quote for 1 Jan 2028.

Del. Reg. (EU) 2026/269 · at 1-in-50 the back-test paid 1966, 2005, not 2018 · premium = model × 2.5–3.5 · prices: assumptions

Tulva

Next March, the snow will warn us again.

This time, it will have a price.

BUILT THIS WEEKEND

SYKE data pipeline | Joint flood model | Index pricing | Live app | QAE on IBM, IQM & VTT Q50

RESOURCES USED

SYKE open data | IBM Marrakesh | IQM Garnet & Emerald (Resonance) | VTT Q50 (LUMI) | Qrisp | Qiskit | AI coding agents

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack

APPENDIX

Hardware runs in detail

Chip	Circuit	Exact	Raw	Calibrated (90 % CI)	Job
IBM Marrakesh	North-cluster pair, 3 qubits, k = 0–2	0.125 %	0.262 %	**0.149 % (0.140–0.157)**	db3tddsvf2bc73ctoprg
IQM Garnet	North-cluster pair, 3 qubits, k = 0–2	0.125 %	0.368 %	**0.164 % (0.147–0.180)**	01a11cc3-7f6c…
IQM Emerald	Snowy / low-snow ratio, same pair, k = 0–2	2.75×	1.3×	**3.05× (2.6–3.6)**	01a11fa8-65a9…
IQM Emerald	14 rivers ≥4 over 1-in-10, 18 q / 44 CZ	10.9 %	–	**13.8 % (readout-corr.)**	01a11cd4-da38…
IQM Emerald	14 rivers ≥2 over 1-in-50, same circuit	4.8 %	–	**16.0 % (fails)**	01a11f9b-c62a…
IQM Emerald	Control: independent circuit, ≥4 over 1-in-10	4.4 %	–	**9.7 % (noise fattens tails)**	01a120f5-ece7…
VTT Q50	Same pair, 3 repeats: not stable	0.125 %	1.48–1.58 %	**0.47 / 0.53 / 0.11 %**	7d20d8ac / a6fb9d52 / 381179fb
VTT Q50	14 rivers ≥4 over 1-in-10, 18 q / 49 CZ	10.9 %	34.2 %	**31.2 % (readout-corr.)**	9cdb0158-7574…

Calibrated = noise-aware MLAE, two known-answer calibration circuits in the same job; bootstrap 90 % intervals. IBM / Garnet intervals exclude exact (+19 % / +31 % bias). Pair = two rivers with identical North-cluster loadings and 1-in-50 trigger. Classical Monte Carlo at the same budget is more precise (±0.004 pp).

Signal kept at k = 1 in the runs shown: Emerald 0.56–0.59, IBM / Garnet 0.77–0.78, VTT Q50 0.12–0.19 (earlier runs 0.65–0.88): the estimate is mostly the calibrated k = 0 circuit. Q50's zero-calibration offset moved 0.002 → 0.006 between repeats (the 5× spread). Not shown: first runs IBM 0.146 % (k = 0–3), Garnet 0.073 % (1 copy).